Have you attempted to answer these questions yet? Do so before scrolling down and compare your thoughts to the key!

Chapter 6 Practice Exam Questions

CHEM 1310 | Spring 2026

1. Indicate whether each statement below is true or false.

   [True False] Heat flow from a colder body to a hotter body is a violation of the first law of thermodynamics.
   [True False] Heat, work, and energy all share the same units.
   [True False] The specific heat of liquid water is equal to the specific heat of solid water (ice).
   [True False] In any process in which a system does work on its surroundings, the internal energy of the system must decrease.

2. Indicate whether each statement below is true or false.

   [True False] Heat flow from a colder body to a hotter body is a violation of the second law of thermodynamics.
   [True False] Heat, work, and energy all share the same units.
   [True False] The specific heat of liquid water is equal to the specific heat of solid water (ice).
   [True False] When a system does work on its surroundings, the internal energy of the system may remain unchanged.

3. Indicate whether each statement below is true or false.

   [True False] Heat flow from a colder body to a hotter body is a violation of the second law of thermodynamics.
   [True False] Heat, work, and energy have different units.
   [True False] The specific heat of liquid water is equal to the specific heat of solid water (ice).
   [True False] When a system does work on its surroundings, the internal energy of the system may remain unchanged.

4. Inside an internal combustion engine, carbon dioxide gas rapidly expands and cools immediately after combustion. Fill in the blanks below to indicate the signs of heat, work, and internal energy change for this process, considering the carbon dioxide gas as the system.

   The sign of work is [positive negative zero] and the sign of heat is [positive negative zero]. The internal energy of the gas [decreases increases does not change].

5. Inside an internal combustion engine, a piston is pushed upward and heats up immediately after combustion. Fill in the blanks below to indicate the signs of heat, work, and internal energy change for this process, considering the piston as the system.

   The sign of work is [positive negative zero] and the sign of heat is [positive negative zero]. The internal energy of the piston [decreases increases does not change].

6. A typical battery for an electric car contains 150 megajoules (MJ) of electrical energy when fully charged. Considering the battery as a thermodynamic system, determine whether each statement below is true or false.

   A. In the process of driving, ΔU for the battery is negative. T
   B. When charging, the battery is doing work on the surroundings. F
   C. In the process of driving, ΔU for the universe (system + surroundings) is positive. F
   D. When the temperature of the battery increases, q is positive. T
   E. Theoretically, all 150 MJ of energy can be converted into work that drives the wheels of the car. F

7. A typical battery for an electric car contains 150 megajoules (MJ) of electrical energy (U) when fully charged. Considering the battery as a thermodynamic system, determine whether each statement below is true or false.

   A. In the process of driving, ΔU for the battery is positive. F
   B. When the battery is charging, the surroundings are doing work on the battery. T
   C. In the process of driving, ΔU for the universe (system + surroundings) is negative. F
   D. When the temperature of the battery increases, q is negative. F
   E. It is not theoretically possible for all 150 MJ of energy to be converted into work that drives the wheels of the car. T

8. A typical battery for an electric car contains 150 megajoules (MJ) of electrical energy when fully charged. Considering the battery as a thermodynamic system, determine whether each statement below is true or false.

   A. In the process of driving, ΔU for the battery is negative. T
   B. When charging, the battery is doing work on the surroundings. F
   C. In the process of driving, ΔU for the universe (system + surroundings) is zero. T
   D. When the temperature of the battery increases, q is negative. F
   E. Theoretically, all 150 MJ of energy can be converted into work that drives the wheels of the car. F